higher education & training

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REPUBLIC OF SOUTH AFRICA

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NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N4

(8080164)

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09:00–12:00

This question paper consists of 6 pages and a formula sheet of 2 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

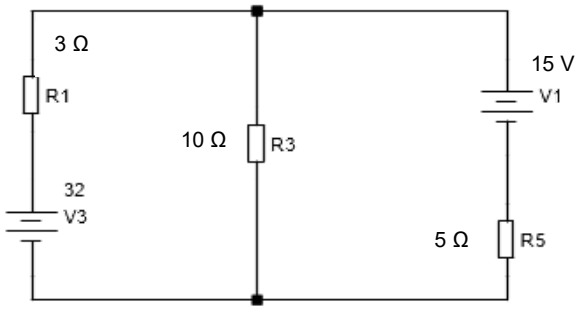
INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 Study the circuit diagram below.



Calculate the current flowing across the 10 Ω resistor using Kirchhoff's law if the current flowing through 3 Ω is I_1, the current flowing across 5 Ω is I_2 and the current flowing across 10 Ω is $I_1 + I_2$. (7)

1.2 Consider the circuit diagram in QUESTION 1.1 and calculate the maximum power across the 10 Ω resistor. (5)

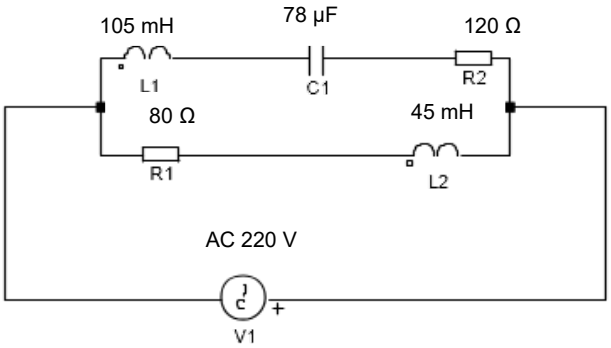
[12]

QUESTION 2

2.1 An LC circuit resonating at 1 500 kHz has a Q of 250.

Calculate the edge frequency of F_L. (3)

2.2 Study the circuit diagram below.



Calculate the total current using j-notation. (8)

[11]

QUESTION 3

3.1 Explain each of the following conditions in an N-type material:

3.1.1 Free electrons

3.1.2 Holes

(2 × 2) (4)

3.2 Calculate the room temperature of a PN-junction diode if the forward resistance is 5,213 Ω which conducts 2 mA. (5)

3.3 What is the other name for a varactor diode? (1)

3.4 Explain the *negative resistance region* in a tunnel diode. (2)

[12]

QUESTION 4

4.1 What is *rectification*? (2)

4.2 Draw a circuit diagram of a full-wave rectifier using a centre-tapped transformer as well as the input and output waveforms. (5)

4.3 Half-wave rectifier output is filtered by an RC filter stage with C = 150 µF and R = 1,2 kΩ. The first capacitor filter develops a 30 V_{dc} with a 4 V_{rms} ripple voltage when supply frequency is 50 Hz.

Calculate the output ripple voltage and the input ripple factor. (4)

[11]

QUESTION 5

5.1 Explain the *cut-off region* of a BJT transistor. (2)

5.2 Explain the position of the bias point for both class A and class B amplifiers. (4)

5.3 Explain each of the following feedback signals:

5.3.1 Series feedback

5.3.2 Shunt feedback

(2 × 2) (4)

5.4 How is the Q-point selected in the load line? (2)

[12]

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QUESTION 6

6.1 An inverting op amp has R_{in} = 3 kΩ and R_f = 6 kΩ.

Calculate the voltage gain and input voltage (V_{in}) if the output voltage (V_o) = 7,3 V. (4)

6.2 A summing amplifier has the following data: R_f = 15 kΩ, V_o = -3,13 V, R_1 = 2 Ω, R_2 = 7 Ω, R_3 = 6 Ω, V_1 = 8,35 V, V_2 = unknown and V_3 = 9,25 V.

Calculate the value of the unknown voltage (V_2). (5)

6.3 Name ONE op amplifier that can be used without feedback. (1)
[10]

QUESTION 7

7.1 Explain each of the following terms:

7.1.1 Thyristor

7.1.2 Holding current of an SCR

(2 × 2) (4)

7.2 Give FOUR disadvantages of an SCR. (4)

7.3 How is an LASCR switched on? (3)
[11]

QUESTION 8

8.1 State TWO points to consider when selecting a transducer. (2)

8.2 Explain each of the following conditions that takes place on a strain gauge:

8.2.1 If NO pressure is applied to the active gauge (2)

8.2.2 If pressure IS applied to the active gauge (4)

8.3 Explain the shaft rotation of a tacho-generator. (3)
[11]

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QUESTION 9

9.1 Give TWO functions of the anode system in electrostatic focussing. (2)

9.2 Explain the function of each of the following parts of a block diagram of a function generator:

9.2.1 Integrator

9.2.2 Sine wave

(2 × 2) (4)

9.3 Signal generators and function generators produce a variety of waveforms for different uses.

Give the type of waveform for each of the following uses:

9.3.1 Transmission of radio signals (1)

9.3.2 Audio frequency generation (1)

9.3.3 Timing circuits in digital electronics (2)

[10]

TOTAL: 100

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FORMULA SHEET

Any applicable formula may be used.

$$\frac{1}{R_T} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \dots \frac{1}{R_n}\right) \qquad R_T = \frac{R_1 R_2}{R_1 + R_2} \qquad V_2 = \frac{R_2}{R_1 + R_2} \times \frac{V_T}{1}$$

$$Z = \sqrt{R^2 + (X_L - X_c)^2} \qquad \text{Cos}\ \theta° = \frac{R}{Z} \quad P = I^2 R \qquad P = \frac{V^2}{R} \qquad P = VI \cos\theta$$

$$P = V\ .\ I \qquad F_r = \frac{1}{2\pi\sqrt{LC}} \qquad Q = \frac{X_L}{R} \qquad OR\ \frac{1}{R}\sqrt{\frac{L}{C}}$$

$$Z = \frac{1}{\sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_c} - \frac{1}{X_L}\right)^2}} \qquad \frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{I_2}{I_1}$$

$$I_t = \sqrt{I_R^2 + (I_c - I_L)^2} \qquad i = I_s\left(e^{\frac{gv}{kT}} - 1\right) \qquad R = \frac{kT}{qi} \qquad V\ .R = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

$$V_{rms} = 0,707\ V_m$$

$$V_{ave} = 0,637\ V_m$$

$$f = \frac{1}{t} \qquad \text{Rate of change} = -\frac{V_{in}}{CR_{in}}$$

$$V_{dc} = 0,318\ V_m \qquad V_{dc} = 0,637\ V_m$$

$$V_{r_{\text{rms}}} = 0,385\ V_m$$

$$PIV = V_m\ OR\ 2\ V_m$$

$$V_{r_{rms}} = \frac{V_r(p-p)}{2\sqrt{3}} \qquad V_{dc} = V_m - \frac{V_r(p-p)}{2}$$

$$r = \frac{V_{r_{rms}}}{V_{dc}} \qquad V_{r_{rms}} = \frac{V_{dc}}{R_L\ 2\sqrt{3}\ FC}$$

$$V_{dc} = V_m - \frac{I_{dc}}{2FC} \qquad r = \frac{I_{dc}}{V_{dc}\ 2\sqrt{3}\ FC}$$

$$V_{r'_{rms}} = \frac{X_c}{\sqrt{R^2 + X_c^2}} \times \frac{V_{r_{rms}}}{1}$$

$$V'_{dc} = \frac{R_L}{R_L + R_S} \times \frac{V_{dc}}{1} \qquad V_{r'_{rms}} = \frac{V_{r_{rms}}}{(2\pi f)^2\ LC}$$

$$R_{in} = \frac{V_{be}}{I_b} \qquad R_{out} = \frac{V_{ce}}{I_c} \qquad R_c = \frac{V_{cc}}{I_c} \qquad V_{out} = R_1 C\ \frac{dv_i}{dt}$$

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Static current gain = $\frac{I_{out}}{I_{in}}$

Dynamic current gain = $\frac{\Delta I_{out}}{\Delta I_{in}}$

$V_{cc} = V_{RC} + V_{ce}$ $V_{ce} = V_{cc} - V_{RC}$ $R = \frac{p\ell}{a}$

$A_p = 10 \log \frac{P_{out/uit}}{P_{in}}$ $A_v = 20 \log \frac{V_{out}}{V_{in}}$ $A_i = 20 \log \frac{I_{out}}{I_{in}}$

Static voltage gain = $\frac{V_{out}}{V_{in}}$

Dynamic voltage gain = $\frac{\Delta V_{out}}{\Delta V_{in}}$

$hie = \frac{\Delta V_{in}}{\Delta I_{in}} = \frac{\Delta V_{be}}{\Delta I_b}$ $V_{ce} =$ constant

$hre = \frac{\Delta V_{in}}{\Delta V_{out}} = \frac{\Delta V_{be}}{\Delta V_{ce}}$ $I_b =$ constant

$hfe = \frac{\Delta I_{out}}{\Delta I_{in}} = \frac{\Delta I_c}{\Delta I_b}$ $V_{ce} =$ constant

$hoe = \frac{\Delta I_{out}}{\Delta V_{out}} = \frac{\Delta I_c}{\Delta V_{ce}}$ $I_b =$ constant

$V_{out} = -\frac{R_f}{R_{in}} \times V_{in}$ $V_{out} = -\left(\frac{R_f V_1}{R_1} + \frac{R_f V_2}{R_2} + \ldots \frac{V_n R_f}{R_n}\right)$

$V_{out} = \left(1 + \frac{R_f}{R_{in}}\right) \quad V_{in}$ $V_{out} = -\frac{1}{CR_{in}} \int V_{in}(t)\, dt$

Boltzmann's constant = $1{,}38 \times 10^{-23}$ J/k

Electron charge = $1{,}6 \times 10^{-19}$ C

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